

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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ABSTRACT. A prepared causal experiment determines both a space of future predictions and a probability law on that space. We prove a resource-transfer theorem in which the finite-memory scale is derived from this acquired law, rather than from an ambient parameter dimension. The hypotheses are certificates on future tests, raw-history mass, valid representative domains, and the propagation of errors introduced by actual updates. An acquisition-weighted transport estimate converts anisotropic small-ball bounds into matching checkpoint and online risk curves, for every positive label budget. Exact waiting transitions introduce no new quantization error. Consequently the constants can remain uniform through arbitrarily long inactive periods, vanishing acquisition probabilities, and changes of acquired rank. Calibration and numerical errors enter before squaring; causal simulation defects enter through the common bounded decision loss, with simulator, controller, and clock states charged separately. We verify the theorem directly for a binary hidden-state filter with continuous reports and for a stratified geometric sensor with absorbing acquisition. The latter has an explicit joint acquisition-memory law through all rank collapses. Classical prediction, quantization, and experiment-comparison results are separated from the acquisition-weighted causal synthesis.

1. THE EXPERIMENTAL PROBLEM

The mother problem is to pass from a physically executable experiment to a quantitative law for the predictions that can actually be retained. There are two distinct obstructions. A set of formally reachable predictions need not receive appreciable mass under the preparation and exploration being used. Moreover, a good codebook at a checkpoint need not admit a good recursive implementation. The first obstruction is probabilistic; the second is causal. Treating either as a dimension count loses the mechanism of the problem.

Our main theorem connects these obstructions by weighting the propagation of quantization errors with the same acquisition probabilities that enter the lower bound. Inactive steps are especially important. When an update preserves a retained representative exactly, rounding need not be repeated. Charging an error at every physical tick would produce a spurious inverse acquisition probability in a system that simply waits. The theorem retains the actual clock while localizing error injection to the updates that require it.

The construction of states from predictions is established background; controlled predictive representations already use future action-observation tests [5]. Squared-loss projection and probability quantization are classical [4]. Recursive quantization of nonlinear filters has a substantial literature [6, 7, 3]. Experiment comparison originates in particular with Blackwell [1]. Our contribution here is not a claim of priority for any of these ingredients. It is the acquired-mass/innovation-transport theorem below, its all-budget rank-uniform geometric consequence, and the two kernel-level verifications. Neither realization is used in the proof of the general theorem.

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1.1. Raw data, strategies, and preparation. A discrete-time experiment consists of standard Borel spaces X_t, A_t, Y_{t+1} , a probability preparation μ on X_0 , and nonnegative normalized kernels

$$(1) \quad K_t(dx', dy, dc \mid x, a).$$

The visible history h_t records actions, reports, and the published nonnegative resource increments c . These increments include failed preparations and physical elapsed time. Positivity means nonnegativity and normalization, not a positive lower bound for every report. Deterministic transitions are permitted. An unknown physical parameter, when present, is included in the hidden state with a specified prior for the Bayesian assertions below. Parameter-uniform experiment comparison will be stated separately.

A legal strategy is a stochastic kernel $\alpha_t(da \mid h_t)$ supported on the declared available actions. All randomness used by the apparatus or algorithm is independent of the hidden preparation unless generated by (1). A stopping rule is a stopping time for the visible filtration. The main risk theorem fixes a deterministic raw-acquisition horizon N and an exploration strategy β independently of the encoder. It does not optimize β . At most N actual calls, including unsuccessful calls, are made. A stopped comparison is always explicitly bounded by N .

Successive kernel integration gives the path law P^α . The complete conditional belief $\pi_t(h_t)$ on the hidden state has a measurable version. We specify versions on a declared admissible history domain; identities on impossible reports are not inferred by division by zero.

1.2. Executable future tests and the quotient. A future test is a legal finite continuation followed by a bounded measurable function of its actual reports. Hidden-state functions qualify only when a terminal measurement producing them is part of the experiment. Tests include determining report events and the continuation interface: legal actions, remaining budget, and every clock value affecting future experiments. They are closed under legal one-step continuation, finite randomization, and the bounded monotone limits needed to determine the designated laws.

For an admissible prepared belief b , let $\mathbf{e}_t(b)$ be its evaluations on a countable determining family of these tests, together with the continuation interface. Equality of $\mathbf{e}_t$ is *predictive equivalence*. We use the following verifiable realization certificate. The admissible belief domains $\mathcal{B}_t$ and the action domains are compact metric; the determining evaluations and the report probabilities are continuous. For continuous reports on a declared compact report domain, there is a common dominating measure and a jointly continuous normalized update on the declared domain. Where the Bayes ratio is used its density denominator is strictly positive. Any extensions at zero denominators are separately verified to be continuous and predictive-fibre-compatible. Alternatively, finite reports with continuous normalized updates suffice. The update is assumed to preserve the declared belief domains. The test family determines the full designated continuation laws. Compactness is a certificate for this class, not an axiom asserting that every experiment has a compact quotient.

Proposition 1.1 (Predictive realization and minimality). *Under this certificate, $S_t = \mathbf{e}_t(\mathcal{B}_t)$ is compact metrizable. It realizes predictive equivalence, and the report probabilities and the normalized update descend to continuous functions and maps on S_t , on the domain where the latter is specified. Any measurable history statistic r_t with respect to which every determining test evaluation and the continuation interface are measurable admits a measurable factor $s_t = f_t(r_t)$ almost surely. A set-theoretic sufficient statistic admits the analogous unique factor on its image, without an automatic Borel assertion.*

Proof. Rescale evaluations into $[-1, 1]$. The countable product is compact metrizable and $\mathbf{e}_t$ is continuous. Its compact image is therefore compact metrizable. Equality on the determining tests implies equality of all designated future laws by induction on continuation depth and a monotone-class argument. In the induction step the numerator and denominator of each conditional test are evaluations of one-step pullbacks; hence equivalent beliefs give equivalent updated beliefs whenever the denominator is positive.

A continuous surjection from a compact space to a Hausdorff space is a quotient map. The same assertion applies after taking the product with the compact action domain and, for continuous reports, the declared compact report domain. Thus every continuous fibre-constant evaluation or update descends continuously. The finite-report case is applied separately to each report. For measurable minimality, apply the measurable factorization lemma to each real evaluation as a $\sigma(r_t)$ -measurable random variable, then combine the countably many factors; on the full-probability set where the resulting vector belongs to S_t , it is the required factor. Outside that set assign a fixed point of S_t . Setwise minimality follows directly from inclusion of the fibres of r_t in predictive fibres. $\square$

Finite exact continuation-interface components can be included as a disjoint compact union. Noncompact report spaces require an independent Borel realization or a verified localization; no universal smooth or standard Borel quotient of an arbitrary equivalence relation is asserted. In the realizations below the quotient is computed explicitly from the raw kernel, which also fixes the conditional versions.

1.3. The common prediction task. At time N , after the label has been retained, an independent index $i \in \{1, \dots, q\}$ is selected with fixed positive weights v_i , $\sum_i v_i = 1$. The corresponding legal terminal probe returns $T_i \in [0, 1]$. The algorithm reports $a_i \in [0, 1]$ and incurs $(T_i - a_i)^2$. Each expectation refers to its own executable probe; a joint counterfactual vector of all probe outcomes is unnecessary. Put

$$(2) \quad p_t(s) = (\sqrt{v_i} \mathbb{E}[T_i | s])_{i=1}^q, \quad B_N = \sum_i v_i \mathbb{E} \text{Var}(T_i | H_N), \quad \nu_N = \text{law}(p_N(s_N)).$$

A varying terminal time uses the corresponding declared probe. The finite scoring menu need not determine the entire predictive quotient. Our assumptions below require only its Lipschitz upper bound globally and a separating lower bound on each acquired chart. Unmeasured interface coordinates cannot be silently charged to this task's lower bound.

A checkpoint encoder reads H_N and retains at most m labels. An online encoder has at most m persistent labels at every time, updates from its previous label and the current input/report, and has no readable history tape. A terminal decoder may depend on the public horizon and on the freshly selected probe index, but not on discarded observations. Shared randomness does not carry observation-dependent state. The risks $R_{\text{cp}}(N, m)$ and $R_{\text{on}}(N, m)$ are infima of the full expected score over these respective classes, under the fixed exploration β . Temporary workspace, numerical input access, and program description are recorded separately in Section 4.

2. THE ACQUIRED-GEOMETRY TRANSFER THEOREM

We state one class theorem. Its assumptions consist of experimental certificates, not an assumed optimal quantization curve. Throughout, J and d_* are finite. Constants may depend on these integers and on the displayed certificate constants, but not on N , the label budget, the positive chart widths, or the acquisition weights.

2.1. Geometric certificates extracted from histories. Let (D, d) be a compact valid predictive-state domain, with $\text{diam } D \leq D_0$, containing the states under consideration and all representatives used by the algorithm. This is a *valid envelope*, not a claim about the support of the acquired law. Updates on D must have the declared physical meaning or a specified conditional version. Time or interface components, if necessary, are included and charged. Suppose the task map satisfies

$$(3) \quad \|p_N(s) - p_N(s')\| \leq Ld(s, s').$$

For each N , disjoint measurable events $E_{N,j}$ partition the actual history law up to a null set. Write $w_{N,j} = P^\beta(E_{N,j})$. Zero-weight components can be omitted. The following conditions are checked

by integrating the raw preparation and kernels, not by choosing a convenient auxiliary geometric measure.

(G) Actual mass and global geometry. There are valid compact component domains $D_j \subset D$ covering D , with dimensions $0 \leq d_j \leq d_*$, and maps

$$\phi_j : Q_j \longrightarrow D_j, \quad Q_j = \prod_{k=1}^{d_j} [-a_{j,k}/2, a_{j,k}/2], \quad a_{j,1} \geq \cdots \geq a_{j,d_j} > 0.$$

We normalize the coordinates so that $a_{j,1} \leq D_0$. Zero widths are removed, not inverted. Dimension zero means a singleton. The maps are onto, and, for constants $b, B > 0$,

$$(4) \quad b\|x - x'\| \leq \|p_N(\phi_j x) - p_N(\phi_j x')\| \leq Ld(\phi_j x, \phi_j x') \leq LB\|x - x'\|.$$

Conditioned on $E_{N,j}$, the state law is $\phi_{j\#}(f_{N,j} dx)$, where

$$(5) \quad 0 \leq f_{N,j}(x) \leq C_g/|Q_j|, \quad \int_{Q_j} f_{N,j}(x) dx = 1.$$

For dimension zero this means unit mass. In particular these are actual acquired masses; f need not be positive everywhere in the envelope. For $J > 1$, the compact score images $p_N(D_j)$ have pairwise distance at least $\Delta > 0$. Constants below may depend on D_0/Δ and LD_0/Δ . The components can be curved; no differentiable structure, covariance, or Fisher matrix is prescribed. The same component envelope and grids must serve all intermediate updates of the chosen implementation. A certificate depending on N is permissible, but a uniform-in- N conclusion requires uniform constants and one stated implementation for each budget.

Set $\psi_j(0) = D_0^2$ and, for integers $k \geq 1$, define

$$(6) \quad \psi_j(k) = \begin{cases} \max_{1 \leq l \leq d_j} \left(\frac{\prod_{i=1}^l a_{j,i}}{k} \right)^{2/l}, & d_j > 0, \\ 0, & d_j = 0. \end{cases} \quad \Psi_N(m) = \min_{\substack{k_j \in \mathbb{N} \cup \{0\} \\ 1 \leq \sum_j k_j \leq m}} \sum_j w_{N,j} \psi_j(k_j).$$

There are finitely many allocations, so the minimum is attained. The zero-allocation penalty retains the cost of discarding an entire acquired component. In the case $J = 1$ it is never needed. Formula (6) is explicit in widths and probabilities obtained from the experiment; it is not a variational definition of optimal distortion.

2.2. Causal certificates. For every allocation (k_j) , finite grids in the domains with $k_j > 0$ and a fixed Borel tie rule define a codebook of at most m representatives. A discarded component is encoded into an existing representative. Write $r_j = \sqrt{\psi_j(k_j)}$. The geometric construction in Lemma 3.2 gives component covering errors at most $C_0 r_j$. The following additional certificate is the causal content.

(I) Innovation-localized implementation. There is a specified parameter-independent update of the representative label, executable from its preceding label, the current command/report, and the declared finite-precision primitives. Along every admitted common command/report path, the exact and representative states satisfy

$$(7) \quad \begin{aligned} e_0 &\leq C_0 \sum_j B_{0,j} r_j + u_0, \\ e_{t+1} &\leq A_{t+1} e_t + C_0 \sum_j B_{t+1,j} r_j + u_{t+1}, \quad e_t = d(s_t, \hat{s}_t). \end{aligned}$$

The nonnegative measurable coefficients $A_t, B_{t,j}$ are determined by the raw experiment and specified update bounds, independently of the grid allocation and precision budget. They may use latent marks for analysis; the implemented machine must not read these marks. The nonnegative u_t bound calibration and numerical errors in the same metric. One sufficient verification is a same-input

Lipschitz bound A_t for the exact update and a rounding error bounded by $C_0 \sum_j B_{t,j} r_j$. An exact hold or grid-preserving transformation is executed without rounding, and then has $B_{t,j} = 0$. An informative overwrite has $A_t = 0$. These checks concern the actual one-step kernel and the given algorithm, not its optimality.

Define empty products as one and set

$$(8) \quad Z_{N,j} = \sum_{t=0}^N B_{t,j} \prod_{i=t+1}^N A_i, \quad U_N = \sum_{t=0}^N u_t \prod_{i=t+1}^N A_i.$$

(T) Acquisition-weighted transport. For all nonnegative real vectors $z = (z_j)$,

$$(9) \quad \mathbb{E}_{P^\beta} \left(\sum_j Z_{N,j} z_j \right)^2 \leq \Gamma_N \sum_j w_{N,j} z_j^2.$$

This is a moment certificate on a positive transport process specified before choosing the memory budget. For example, the positive semidefinite inequality $\mathbb{E}[Z_N Z_N^\top] \preceq \Gamma_N \text{diag}(w_N)$ suffices. The weaker inequality on the nonnegative cone in (9) is all that is used. When a component has zero acquired mass, its transport contribution must vanish almost surely. Verifying this point prevents unacquired geometry from entering an average risk law.

(C) Report continuity and effective primitives. On common continuation interfaces the conditional report kernels obey

$$(10) \quad \|g_t(\cdot \mid s, a) - g_t(\cdot \mid s', a)\|_{\text{TV}} \leq \kappa_t d(s, s').$$

A calibrated numerical implementation may additionally incur a same-state report error ζ_t . The grid descriptions, update routines, readouts, and finite allocation search in (6) are specified by finite programs with declared calibration/numerical input access. Exact algebraic width and weight data suffice for the finite allocation comparisons. A certified constant-factor allocation is also sufficient, changing only the upper constant; arbitrary comparisons of unspecified real oracles are not assumed effective. A metric covering argument alone does not prove bit computability; this effective certificate is required for the word “implementable.” The finite algorithms in Section 5 supply it explicitly. A terminal readout approximation contributes at most v_N^{num} in Euclidean score norm.

Theorem 2.1 (Causal acquired-geometry and resource transfer). *Fix the raw experiment, preparation, exploration, task, and horizon N . Under the predictive realization certificate and (G), (I), (T), (C), there are $c, C > 0$ with the dependence specified above such that, for every integer $m \geq 1$,*

$$(11) \quad B_N + c\Psi_N(m) \leq R_{\text{cp}}(N, m) \leq R_{\text{on}}(N, m) \leq B_N + \left(C\sqrt{\Gamma_N \Psi_N(m)} + L\|U_N\|_{L^2} + v_N^{\text{num}} \right)^2.$$

The online upper is attained as an inequality by the specified finite representative machine, with an allocation minimizing (6). In particular, with exact arithmetic and calibration, and $\sup_N \Gamma_N < \infty$,

$$(12) \quad R_{\text{cp}}(N, m) - B_N \asymp R_{\text{on}}(N, m) - B_N \asymp \Psi_N(m)$$

uniformly over the certified horizons, positive widths, and acquisition weights. Widths may vanish and the acquired rank may change, provided the surviving-coordinate certificates have the same constants. There is no inverse of a vanished width in the conclusion.

For a common bounded decision loss of range at most one, the same implementation yields a finite-horizon report-law allowance

$$(13) \quad \min \left\{ 1, \sum_{t < N} \mathbb{E}[\kappa_t e_t + \zeta_t] \right\},$$

whenever (I) holds on the prefixes needed for the comparison. A common visible stopping time $\sigma \leq N$ inserts $\mathbf{1}_{\{t < \sigma\}}$ in this sum. Additional causal simulators compose this allowance with their

certified defects, and compose their resource maps as in Theorem 4.1. None of these assertions compares two different full-history oracle baselines.

The acquisition term can be made explicit without inventing a universal power of N . For a specified common ideal baseline b_* with $B_N \geq b_*$, put $I_N = B_N - b_*$. Then the exact implementation has

$$(14) \quad R_{\text{on}}(N, m) - b_* \asymp I_N + \Psi_N(m).$$

The quantity I_N must itself be computed or bounded from the actual acquisition kernel. Section 5.2 computes it exactly as a failed-acquisition probability times a target variance. Equation (14) is not a claim that every experiment has the same statistical exponent.

3. PROOF OF THE TRANSFER THEOREM

Lemma 3.1 (Checkpoint projection). *For the task (2),*

$$(15) \quad R_{\text{cp}}(N, m) = B_N + \inf_{c_1, \dots, c_m \in \mathcal{C}} \int \min_l \|x - c_l\|^2 \nu_N(dx), \quad \mathcal{C} = \prod_i [0, \sqrt{v_i}].$$

Randomized encoders and decoders do not improve this infimum.

Proof. For each executed probe, condition on H_N and on any independent encoder randomization. The cross term in

$$(T_i - a_i)^2 = (T_i - \mathbb{E}[T_i | H_N])^2 + (\mathbb{E}[T_i | H_N] - a_i)^2 + 2(T_i - \mathbb{E}[T_i | H_N])(\mathbb{E}[T_i | H_N] - a_i)$$

has conditional expectation zero. Sum with weights v_i . For a retained label, replace a randomized decoder by its conditional mean; convexity cannot increase the risk. For fixed centers, replace a randomized label by a nearest center with the first-index tie rule. This is measurable. Conversely, this rule is a legal checkpoint encoder and attains its codebook distortion. Independent shared random seeds only average the costs of such deterministic codebooks, so cannot lower their infimum. Projection of a center to the box $\mathcal{C}$ is nonexpansive relative to all points of $\mathcal{C}$. Compactness gives attainment of the codebook minimum, although only its infimum is needed. $\square$

Lemma 3.2 (Actual anisotropic mass and allocation). *Under (G), the distortion in (15) lies between $c\Psi_N(m)$ and $C\Psi_N(m)$ for every $m \geq 1$. Each component with $k \geq 1$ has a k -point valid-state cover of radius at most $B\sqrt{d_*\psi_j(k)}$. The conclusions hold with the same constants when any collection of widths is removed at zero.*

Proof. Fix a component of positive dimension d and write $h = \sqrt{\psi_j(k)}$. Put $R = 2h$. For coordinates with $a_i > R$, divide the interval into $n_i = \lceil a_i/R \rceil$ equal pieces; for the other coordinates take one piece. If there are l active coordinates, they are the first l , and

$$\prod_i n_i \leq \prod_{i=1}^l \frac{2a_i}{R} = \frac{\prod_{i=1}^l a_i}{h^l} \leq k.$$

The assertion also holds when $l = 0$. Every coordinate interval has length at most R . Its midpoint product grid has Euclidean covering radius at most $\sqrt{d}h$, and (4) gives the valid-state radius. A point component takes one label and has zero radius. An omitted component can use any retained representative with error at most D_0 . Minimizing the allocation proves the upper bound, including the cases $m < J$ and atoms of arbitrary weight.

For the lower bound, a score ball of radius r has preimage of Euclidean diameter at most $2r/b$. If it meets the component, its projection on any one coordinate is contained in an interval of length at most $4r/b$. Integrating the actual density in (5) over the remaining coordinates gives, for $1 \leq l \leq d$,

$$(16) \quad P^\beta \{p_N(s_N) \in B(c, r) \mid E_{N,j}\} \leq C_g \frac{(4r/b)^l}{\prod_{i=1}^l a_i}.$$

The center c need not lie on the image. This is a bound on the actual experiment, not on a fictitious uniform distribution.

If $J = 1$, all m centers can be used in the union bound. For a fixed l , choose $r = c_l(\prod_{i=1}^l a_i/m)^{1/l}$ so that the union has probability at most $1/2$. The expected squared error is at least $r^2/2$. Maximize over l .

If $J > 1$, assign each decoder center to a component whose score image is at distance less than $\Delta/3$, if such a component exists. Separation makes this assignment unique. Let k_j be the number assigned. For $r < \Delta/3$, unassigned or differently assigned centers do not cover any point of component j . When $k_j \geq 1$, use (16) and the preceding radius, capped at $\Delta/4$. Since $\psi_j(k_j) \leq D_0^2$, the cap changes the lower-bound constant only by a factor depending on D_0/Δ , b , and C_g . For $k_j = 0$, the component's distortion is at least $\Delta^2/9$, hence at least a fixed multiple of $\psi_j(0)$. At a point with $k_j \geq 1$ the required lower bound is zero. Multiply each component bound by its actual weight and sum. We have $\sum_j k_j \leq m$. If this sum is zero, give one component a label: its ψ_j cannot increase, since $a_{j,1} \leq D_0$. Thus the sum is bounded below by a constant times the minimum in (6). This proves the lower bound for arbitrary decoder centers and all budgets.

Only surviving coordinates were used. No constant in the covering or small-ball argument contains a_i^{-1} except within the displayed physical volume, which is eliminated in the final scale. This proves the rank-uniform assertion. $\square$

Lemma 3.3 (Innovation transport). *Under (I), for every admitted path,*

$$(17) \quad e_N \leq C_0 \sum_j Z_{N,j} r_j + U_N.$$

Consequently (T) implies

$$\|e_N\|_{L^2} \leq C_0 \sqrt{\Gamma_N \sum_j w_{N,j} r_j^2} + \|U_N\|_{L^2}.$$

Proof. Induct in (7), distributing each nonnegative multiplicative coefficient. This gives precisely (8) and (17). Minkowski's inequality and (9), evaluated at $z_j = r_j$, give the second assertion. No independence of the coefficients is used. $\square$

Proof of Theorem 2.1. Lemma 3.1 and Lemma 3.2 give the lower bound for every checkpoint encoder. The terminal output of any online encoder is a checkpoint encoder, so $R_{\text{cp}} \leq R_{\text{on}}$. Choose a minimizing allocation in (6) and its valid grids. Execute the label update in (I). By (3), its score error has L^2 norm at most $L\|e_N\|_{L^2} + v_N^{\text{num}}$. The conditional projection identity is valid for this online encoder as well. Apply Lemma 3.3, and absorb LC_0 into C . This proves (11). Its exact-arithmetic specialization gives (12); adding the same nonnegative I_N yields (14).

For path comparison, run the exact and representative report experiments with the same controller at identical visible prefixes. At such a prefix the action kernels coincide, the retained label is the same function of the prefix, and the next-report kernels differ by at most $\kappa_t e_t + \zeta_t$. Maximal coupling of the next report is possible on standard Borel spaces. The common-prefix subprobability law is dominated by the exact prefix law. Hence the probability of a first disagreement at step $t+1$ is at most $\mathbb{E}_{P^\beta}[\kappa_t e_t + \zeta_t]$, or the corresponding expectation for the controller actually being compared. Summing proves (13). Append the common label update and resource readout; these do not increase the discrepancy. After the common stopping decision, both processes are absorbed, which inserts the active-step indicator. To use the numerical bound from (T) for a different controller, that controller must have its own certificate; the fixed- β certificate is not automatically uniform over policies. Composition and common-loss transfer are proved in the next section. $\square$

3.1. How to verify the transport certificate.

Proposition 3.4 (Three kernel-level mechanisms). *The following are sufficient verifications of (T).*

(a) *With one component of weight one, $A_t \leq \rho < 1$, $B_{t,1} \leq 1$, and $B_{0,1} \leq 1$, one may take $\Gamma_N \leq (1 - \rho)^{-2}$.*

(b) *Suppose each update is either an exact hold with $A_t = 1$, $B_{t,1} = 0$, or an active step with $A_t \leq \rho < 1$, $B_{t,1} \leq 1$. Then the same constant works, without a lower bound on the probability or frequency of active steps. Here $w_{N,1} = 1$; an initial rounding is permitted.*

(c) *Suppose the process starts at one known unresolved point, waits there, and at a first successful acquisition moves into component j , thereafter holding. The update on a successful report overwrites the old prediction, including its conditional version at an otherwise impossible report. Let $B_{0,0} = 1$, and let $B_{t,j} = 1$ only at the acquisition into j . Then*

$$Z_{N,0} = \mathbf{1}_{\{\text{no acquisition by } N\}}, \quad Z_{N,j} = \mathbf{1}_{\{\text{acquired component } j \text{ by } N\}}.$$

For the corresponding actual component weights, (9) is an equality with $\Gamma_N = 1$.

Proof. In (a), sum a geometric series. In (b), delete the exact holds from the error recurrence, not from the raw acquisition or time account. The resulting pathwise series is the same geometric series indexed by active events. In (c), failure and post-acquisition holds have $A = 1, B = 0$, while the acquisition has $A = 0$ and the appropriate $B = 1$. Thus the initial contribution is killed by the first acquisition and the new contribution persists. The displayed indicators are disjoint and exhaustive. Squaring their weighted sum and taking expectation gives the assertion. $\square$

Mechanism (b) permits arbitrarily long metastable waiting periods. It is different from replacing the raw horizon by an uncharged number of successful reports. Mechanism (c) additionally preserves the small acquired mass in the upper bound, rather than paying the full geometric cost on paths that never acquire it. More general nonuniform stability is covered whenever the positive moment inequality (9) is separately established; it is not asserted from a negative average logarithmic Lipschitz exponent alone. The relation to average contraction theory [2] is thus precise: here the required object is a forced, acquisition-weighted second moment, not merely convergence of unforced iterates.

Corollary 3.5 (Intersecting strata). *Remove the separation assumption from (G), allow the score images of the J component charts to intersect, and let the raw preparation carry a latent component mark with probabilities $w_{N,j}$. The mark need not be available to the algorithm or retained by the quotient. Suppose the conditional laws and the other certificates remain valid, with chart rounding certified conditionally on the raw mark. Then (11) and (12) hold for every $m \geq 2J$, with constants also depending on J , and without any inverse distance between strata.*

Proof. The global upper cover is still the union of the component grids. Encode into a nearest valid representative in d ; its error on component j is no greater than that component grid's error. Duplicate representatives may be merged. The algorithm does not need the latent mark.

For the lower bound apply (16) with all m decoder centers to each positive-dimensional conditional law. This gives distortion at least $c \sum_j w_{N,j} \psi_j(m)$; point components contribute zero. Set $k = \lfloor m/J \rfloor \geq 1$. Directly from (6), for $m \geq k \geq 1$,

$$\psi_j(k) \leq (m/k)^2 \psi_j(m) \leq (2J)^2 \psi_j(m).$$

The allocation of k labels to every component is feasible. Thus $\Psi_N(m) \leq (2J)^2 \sum_j w_{N,j} \psi_j(m)$. Combining the last two bounds proves the claimed lower, and the remaining causal proof is unchanged. In particular neither smoothness at an intersection nor identifiability of the latent mark was used. $\square$

4. CAUSAL MORPHISMS AND JOINT ERROR BUDGETS

A causal morphism $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ uses a source experiment to implement a target experiment. Its data are a parameter-independent nonanticipating report transformer, a target-to-source strategy

lift, internal simulator state, and explicit correspondences for legal actions, task readout, clock, and costs. Randomization is permitted; access to the unknown parameter or future reports is not. For a parameterized family, its defect is

$$(18) \quad \epsilon_\Gamma = \sup_{\lambda, \alpha \in \mathcal{A}_{\mathfrak{F}}} \|\text{law}_\lambda^\mathfrak{E}(\Gamma; \mathbb{L}_\Gamma \alpha) - P_{\mathfrak{F}, \lambda}^\alpha\|_{\text{TV}}, \quad \|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|.$$

The law includes the common loss-relevant target readout and published cost/clock observations. The supremum is only over the stated horizons and strategy class. Taking the infimum over a declared simulator class gives its causal deficiency; a particular simulator supplies an upper certificate, not an optimality assertion. The fixed-preparation version omits the parameter supremum.

The resource vector is $(N, m, R, D_c, Q, W, P, b_{\text{cal}}, \tau)$. Here m is the representative label count, R the simulator state count, D_c the controller state count, Q an internally stored clock count, W temporary workspace in bits, P program/readonly description length, b_{cal} calibration precision, and τ physical elapsed time. An externally supplied visible clock is declared as such; it is not an unacknowledged free memory tape. Acquisition and time budgets are pathwise unless explicitly marked expected or tail budgets. The raw cost of any calibration pilot is included in N .

Theorem 4.1 (Resource and common-risk composition). *Let $\Gamma_1 : \mathfrak{E} \rightarrow \mathfrak{F}$ and $\Gamma_2 : \mathfrak{F} \rightarrow \mathfrak{G}$ have compatible action, task, clock, and strategy interfaces. Suppose their defect bounds are uniform over the lifted strategy classes. Their serial composition is causal, parameter independent, and has defect at most $\min(1, \epsilon_1 + \epsilon_2)$. Its resource map is $\rho_1 \circ \rho_2$.*

In particular a representative machine of m labels, a controller of D_c states, a simulator of R states, and an internal clock of Q states have an implementation with at most mD_cRQ persistent states. If a target raw call requires at most n_Γ source calls, and a pilot costs n_{pil} calls, the source acquisition budget is at most $n_{\text{pil}} + n_\Gamma N$. Corresponding physical times are added according to the actual protocol. Workspaces and descriptions must be accounted for independently; the state-product formula is not a bound on them.

For a common loss $0 \leq \ell \leq 1$ and a target resource vector b ,

$$(19) \quad \mathcal{R}_\mathfrak{E}(\rho_\Gamma(b)) \leq \mathcal{R}_\mathfrak{F}(b) + \epsilon_\Gamma.$$

For the prediction task of Theorem 2.1, a source implementation consequently satisfies

$$(20) \quad \mathcal{R}_\mathfrak{E}(n_{\text{pil}} + n_\Gamma N, mD_cRQ) \leq B_N + \left(C\sqrt{\Gamma_N \Psi_N(m)} + L\|U_N\|_{L^2} + v_N^{\text{num}} \right)^2 + \epsilon_\Gamma + p_{\text{bad}},$$

when the displayed deterministic approximation certificates fail only on an event of probability at most p_{bad} . If report approximation is a separate causal step, its bounded-horizon defect from (13) is added once, not counted again inside ϵ_Γ .

Proof. Interleave the two simulators in causal order. Their joint internal state, with the target controller and clock, is the Cartesian product just counted. Legal actions and readout/clock correspondences compose by their hypotheses. At the path-law level, applying the second nonanticipating kernel to the first simulation cannot increase total variation. Insert the exact intermediate experiment and use the triangle inequality. Resource maps compose in the same execution order; a pathwise acquisition bound composes pathwise, not merely in expectation.

For any target algorithm, execute it through the simulator. The expectation of a $[0, 1]$ -valued common loss changes by at most the total variation defect, by the layer-cake formula. Infimize over target algorithms to obtain (19). Apply the upper bound of Theorem 2.1 to obtain (20). On a bad-certificate event the loss is at most one; on its complement use the valid error bound. This adds at most p_{bad} , without claiming that conditioning on the good event preserves the original acquired law. $\square$

Calibration and numerical update errors are added in u_t before the products in (8); their propagated root-mean-square contributions add before squaring. State counts multiply, so their

ideal label-bit accounts add as logarithms. Acquisition and physical-time costs use their actual protocol maps. An upper bound with $m = \lfloor M/(D_c R Q) \rfloor$ applies only when this integer is positive. A matching law in the *total* persistent budget M follows with constant overhead only if D_c, R, Q are uniformly bounded, or if a separate lower bound certifies the same mandatory state cut. There is no unproved reverse memory monotonicity.

4.1. Which error terms have matching converses? An allowed tolerance is not an unavoidable error. A perfectly calibrated instance belongs to every larger tolerance class. Similarly, a simulator with zero defect satisfies any positive defect allowance. Thus the following converse is formulated for an explicitly specified robust class, rather than silently inserted into (12).

Proposition 4.2 (Attainable robust error floors). *The following common-task subexperiments give sharp error orders.*

(i) *An unobserved target probability is either $1/2 - h$ or $1/2 + h$, with equal prior probabilities, $0 \leq h \leq 1/4$. Every calibration report before the terminal forecast is identical in the two cases. The terminal report is Bernoulli with that target probability. Relative to the oracle that knows the sign, the minimum excess square score is h^2 .*

(ii) *A hidden fair bit Z is observed exactly in a target experiment. A source experiment instead reports Z with probability $1 - \theta$ and an erasure symbol with probability θ , independently of Z . The task is to predict a terminal report equal to Z under square loss. With enough labels to retain the source report, its minimum risk is $\theta/4$. A parameter-independent simulator that fills an erasure by a fair bit has parameter-uniform defect $\theta/2$ from the exact-bit target experiment, when the hidden parameter is $Z \in \{0, 1\}$ and the terminal readout is included.*

Consequently, for a declared class containing these alternatives and a geometry subexperiment of Theorem 2.1, a uniformly available upper bound of order $I_N + \Psi_N(m) + h^2 + \epsilon$ is matched from below up to a constant, provided all subexperiments use the same loss normalization and the compared quantity is the same specified excess risk in each member. Witnesses for the four nonnegative terms may be different members of this one class; no single simultaneous hardest member is required.

Proof. In (i), the posterior mean is $1/2$; the projection identity gives the excess $\mathbb{E}(p - 1/2)^2 = h^2$. It is attained by the constant forecast. In (ii), the posterior mean is 0 or 1 on a nonerased report and $1/2$ on erasure. Its conditional variance is zero or $1/4$, so the Bayes risk is $\theta/4$. At either fixed Z , filling an erasure produces the wrong report with probability $\theta/2$; the joint terminal target still equals Z , proving the claimed defect. Finally a supremum over a class dominates each subexperiment's lower bound, and the maximum of four nonnegative numbers is at least one quarter of their sum. The last assertion is conditional on the stated common-class embedding and on its uniform upper, not an assertion that arbitrary physical families contain these alternatives. $\square$

The calibration witness also describes compulsory finite-precision information loss when two distinct physical probabilities yield the same numerical input. It does not give a lower bound merely because an implementation *may* use inaccurate arithmetic. A source lower bound cannot be transported by a forward simulator alone. It requires a source witness, or a reverse morphism with its own resource certificate. These distinctions keep (20) within the same task metric.

5. TWO RAW-KERNEL REALIZATIONS

5.1. A controlled binary hidden-state filter. Let $X_t \in \{0, 1\}$ have transition probability $q_0 \in (0, 1/2)$ of flipping at each raw call. The preparation is $P(X_0 = 1) = 1/2$. A command selects a from a fixed finite subset of $[a_-, a_+] \subset (0, 1)$. Conditional on the new hidden state, the report $y \in [-1, 1]$ has density

$$(21) \quad g(y \mid X_{t+1} = x, a) = \frac{1}{2}(1 + a(2x - 1)y).$$

Each call costs one acquisition and one unit of physical time. The exploration may be any specified causal command rule, fixed independently of the encoder. A terminal probe reads X_N and ends the experiment. Further filtering reports and this terminal probe are the designated future tests.

Theorem 5.1 (Filtering realization). *For this raw-kernel class, the predictive quotient is the scalar conditional probability $p_t = P(X_t = 1 \mid H_t)$, with the terminal/continuation interface. It has a common compact invariant interval after initialization. For all $N \geq 1$ and $m \geq 1$,*

$$(22) \quad R_{\text{cp}}(N, m) - B_N \asymp R_{\text{on}}(N, m) - B_N \asymp m^{-2}, \quad B_N = \mathbb{E}[p_N(1 - p_N)].$$

The constants depend only on q_0, a_-, a_+ , not on N or on the specified command rule. The upper has a finite recursive implementation. A uniform update/numerical error u in log-odds distance contributes at most $u/(2q_0)$ to the propagated state error. The constants in this statement are not asserted uniform as $q_0 \downarrow 0$.

Proof. Direct normalization of (21) gives, with $r = q_0 + (1 - 2q_0)p$,

$$(23) \quad f_a(y \mid p) = \frac{1}{2}[1 + ay(2r - 1)], \quad F_{a,y}(p) = \frac{r(1 + ay)}{1 + ay(2r - 1)}.$$

All denominators are at least $1 - a_+$. Conditional laws of all future report words and the terminal probe are functions of p by induction. Conversely the terminal probe has mean p , so different p are predictively different. This proves quotient identification and minimality without assuming a covariance or information matrix.

Let

$$c_- = \frac{q_0(1 - a_+)}{1 + a_+(1 - 2q_0)}, \quad c_+ = 1 - c_-.$$

The interval $D = [c_-, c_+]$ contains $1/2$ and is invariant, because the prediction r always lies in $[q_0, 1 - q_0]$ and the update is increasing in r and ay . Use the metric $d(p, p') = |\text{logit}(p) - \text{logit}(p')|$. On D it is equivalent to Euclidean distance, with constants depending only on the displayed parameters. The map $p \mapsto p$ is the task map, and its inverse-coordinate Lipschitz constants follow from the bounded derivative $1/[p(1 - p)]$. This provides a one-dimensional global chart and an effective valid domain.

For update stability, put $v = p(1 - p)$ and $\rho = 1 - 2q_0$. The derivative of the predicted log-odds with respect to previous log-odds equals

$$\frac{\rho v}{q_0(1 - q_0) + \rho^2 v} \leq \rho.$$

Indeed $v \leq 1/4$ and $1 - \rho^2 = 4q_0(1 - q_0)$, so the denominator is at least v . The observation then adds $\log[(1 + ay)/(1 - ay)]$, independently of the previous state. Thus every same-report update is ρ -Lipschitz in d . Proposition 3.4(a) gives $\Gamma_N \leq (2q_0)^{-2}$, including initialization rounding.

It remains to verify actual acquired mass. For each preceding history and command,

$$(24) \quad \frac{\partial F_{a,y}(p)}{\partial y} = \frac{2ar(1 - r)}{[1 + ay(2r - 1)]^2} \geq c_y := \frac{2a_-q_0(1 - q_0)}{(1 + a_+)^2} > 0.$$

The conditional density of y is at most $(1 + a_+)/2$. Change of variables therefore gives a conditional density of p_{t+1} at most $(1 + a_+)/(2c_y)$ on its conditional interval and zero elsewhere. Mixtures over the actual histories and commands preserve this upper bound. Its integral is one. Consequently (5) holds on the fixed interval D at every $N \geq 1$, with C_g this bound times $|D|$. No uniform posterior law has been inserted. Equations (4) and (5) give $\Psi_N(m) = |D|^2/m^2$.

For report continuity, direct integration of the density difference gives

$$\|f_a(\cdot \mid p) - f_a(\cdot \mid p')\|_{\text{TV}} = \frac{a\rho}{2}|p - p'| \leq \frac{a_+\rho}{8}d(p, p').$$

The terminal Bernoulli kernel has total variation $|p - p'| \leq d(p, p')/4$. Hence all loss-relevant reports satisfy (C). An equally spaced rational grid in p , rational update (23), and nearest-grid threshold comparisons give an explicit implementation. Approximate input and arithmetic with certified absolute errors are absorbed into u_t ; denominator and log-odds conversion bounds above translate them to the common metric. For rational calibration, interval arithmetic terminates to any prescribed positive error. The persistent datum is the grid index; a bounded-precision register for the current report and arithmetic is temporary, not a retained real state. Summing the geometric error series gives the last error assertion. Apply Theorem 2.1 to conclude. $\square$

This is a repeatedly updated nonlinear filter, not a once-prepared finite measurement ensemble. The acquired law is generated by the actual observation density at every positive time. Formula (22) concerns prediction along the selected exploration, not the unrestricted optimal information-gathering controller.

5.2. Singular geometric acquisition with an absorbing report. We next construct a class with no strict contraction during waiting and with genuinely intersecting, different-dimensional acquired supports. Fix finitely many compact boxes Q_j and maps

$$z_j : Q_j \longrightarrow [1/4, 3/4]^d$$

that are bi-Lipschitz with uniform constants in physical Euclidean coordinates. They may be nonlinear and their images may intersect. A dimension-zero box is a point. Prepare a latent mark J_0 with probabilities π_j and, conditionally on $J_0 = j$, a point V uniformly on Q_j . The hidden target is $Z = z_{J_0}(V)$. After preparation the raw hidden state is (Z, A_t) ; the latent mark is retained only on the preparation probability space for analyzing its pushforward law. The mark and any zero-width source coordinates are not supplied to the observer.

The apparatus has an unresolved/absorbed flag $A_t \in \{0, 1\}$, initially zero. At each raw attempt, when unresolved, an independent coin with success probability $s \in [0, 1]$ is tossed. Failure reports $\perp$ and changes nothing. Success reports Z and sets the flag to one. Thereafter every raw attempt reports $\perp$ and preserves Z and the flag. All attempts, including these holds, cost one acquisition and one unit of physical time. The action “attempt” is available in both modes, so no unrecorded action-legality bit is assumed. The terminal menu consists of a flag probe returning A_N , and d binary probes with success probabilities Z_i , each executed only once as a terminal experiment. Give these $d + 1$ probes equal weights.

After failure while unresolved, the hidden target distribution is unchanged: failure probability is independent of Z . After a successful report the future target probabilities are exactly the reported Z , whether or not the latent mark is identifiable. Thus the predictive quotient is the unresolved point o , together with the *glued* set of absorbed predictions $z \in \bigcup_j z_j(Q_j)$. At an intersection the mark is quotiented out, not secretly retained. The flag probe distinguishes o from an absorbed state even when their coordinate means coincide.

Set

$$(25) \quad b_* = \frac{1}{d+1} \sum_{i=1}^d \mathbb{E}[Z_i(1 - Z_i)], \quad V_* = \frac{1}{d+1} \sum_{i=1}^d \text{Var}(Z_i), \quad h_N = (1 - s)^N.$$

For $N = 0$ take $h_0 = 1$, including $s = 1$. Let $\Psi^{\text{acq}}(m)$ be (6) for the absorbed component charts with weights π_j . Widths there are the actual physical chart widths with the fixed task-weight scaling absorbed into the chart constants.

Theorem 5.2 (Stratified acquisition law through rank collapse). *For every $s \in [0, 1]$, $N \geq 0$, and $m \geq 2(J + 1)$,*

$$(26) \quad R_{\text{cp}}(N, m) - b_* \asymp R_{\text{on}}(N, m) - b_* \asymp h_N V_* + (1 - h_N) \Psi^{\text{acq}}(m).$$

The constants depend only on the finite chart and task constants, not on s, N , positive widths, or the distance between intersecting strata. When positive widths vanish, zero coordinates are removed from the predictive quotient and the same assertion holds, including point components. If all component score images (including the unresolved point) are separated, Theorem 2.1 gives the all-budget version using the full mixture allocation (6), for every $m \geq 1$.

The online construction has $\Gamma_N = 1$ in the full-mixture transport certificate. It retains no success count or latent component mark. Finite-precision successful reports with metric error at most η give a root-mean-square prediction allowance at most $C\sqrt{1 - h_N\eta}$, when all hold operations are exact.

Proof. The preparation is the pushforward of the specified box mixture, and the transition is a normalized nonnegative Borel kernel on the compact target space with its two flags. The admissible belief domain is the unresolved prepared law together with the point beliefs at absorbed targets. It is a disjoint compact union. All determining future-test evaluations are continuous on it; the explicit overwrite/hold versions below also verify the zero-denominator extension clause of Proposition 1.1. The probability of N consecutive failures is exactly h_N . Conditional on success by N , the target law is still the prepared mixture, because all acquisition coins are independent of it. The acquired predictive law is therefore

$$(27) \quad h_N \delta_o + (1 - h_N) \sum_j \pi_j(z_j)_\# \text{Unif}(Q_j),$$

with the absorbed flag coordinate included in the score. In particular the formal union of all prepared values does not have weight one before it is acquired.

The conditional terminal variance of the flag is zero. On an absorbed history the coordinate variance is $Z_i(1 - Z_i)$. On an unresolved history it is $\mathbb{E}[Z_i(1 - Z_i)] + \text{Var}(Z_i)$, by total variance. Averaging proves

$$(28) \quad B_N = b_* + h_N V_*.$$

Conditioned on the raw component mark, the acquired charts have exactly uniform physical box densities. Their bi-Lipschitz constants give (G). Their union can have intersections, different local dimensions, and atomic mass. All these properties follow from the raw preparation and report map.

Equip the quotient with Euclidean distance on its weighted score vector. Report continuity holds as follows. Between two absorbed states the next-attempt kernels are identical, namely $\delta_\perp$. Between unresolved and absorbed modes, the variation distance is at most one, while the flag-score distance is $1/\sqrt{d+1}$; a finite global Lipschitz constant follows. Terminal coordinate kernels are Bernoulli and their variation distance is $|Z_i - Z'_i|$; the flag kernel is bounded by the same mode-separation argument. There is no assertion that distinct Dirac reveal channels are TV-continuous: the reveal action is used only from the single unresolved quotient point. On a successful report define the predictive update to overwrite with that reported absorbed state from every previous representative, including the conditional version at a null report in the absorbed mode. On $\perp$, it is the identity.

The machine initially encodes o . At a success it chooses a nearest representative in the union codebook, using the reported physical coordinates; on every failure or later hold it leaves the label unchanged. It never reads J_0 . Conditional on any raw mark j , the success-rounding error is bounded by that chart's cover. Therefore the coefficients are exactly those of Proposition 3.4(c), with the unresolved point as component zero. This proves $\Gamma_N = 1$, irrespective of how many holds occur. It also proves the finite-precision allowance: a single error of size $C\eta$ is introduced on an event of probability $1 - h_N$, and none on the complementary event.

For completeness one can see the separated oracle and acquisition terms directly. Lemma 3.1, conditioned on the success event and its latent mark, and the proof of Corollary 3.5 give an excess of at least $c(1 - h_N)\Psi^{\text{acq}}(m)$, even if the decoder also uses centers for the unresolved point. For the upper use one exact unresolved label and $m - 1$ absorbed grid labels. Allocation scaling gives $\Psi^{\text{acq}}(m - 1) \leq C_J \Psi^{\text{acq}}(m)$ for $m \geq 2(J + 1)$: allocating $\lfloor (m - 1)/J \rfloor$ to every chart and using

the estimate in Corollary 3.5 proves it. The finite machine just constructed then incurs at most $C(1 - h_N)\Psi^{\text{acq}}(m)$. Add (28). At $s = 0$ or $N = 0$ the success term is zero and a single known forecast attains the acquisition term. With zero surviving widths all component charts are points and sufficiently many labels give zero acquired distortion, as the formula states.

For effective implementation, take algebraic calibrated widths and weights, and chart maps with certified finite-precision evaluation. The finite allocation minimum is an algebraic comparison problem. A nearest representative can be selected to within absolute distance η by comparing approximated distances; this adds at most 2η to the best covering radius. No exact comparison of arbitrary input reals is required. A finite readonly table of approximated representatives or a grid-generation program, with separately charged description and temporary workspace, suffices. Thresholds are compared in physical coordinates; no uniform error estimate divides by a collapsing width. This proves the claimed implementation statements. $\square$

A concrete singular family. In two coordinate dimensions take equal mixture weights on

$$z_1(u) = (1/2 + u, 1/2), \quad |u| \leq a/2, \quad z_2(v, w) = (1/2 + v, 1/2 + w), \quad |v| \leq b/2, |w| \leq c/2,$$

where $0 \leq a, b, c \leq 1/4$. The actual acquired law has a one-dimensional singular component on a line and a two-dimensional component on a rectangle. The former lies inside the latter when their horizontal ranges overlap. If $c = 0$, the rectangle becomes a line; if also $b = 0$, it becomes an atom. Before acquisition the support consists only of the unresolved predictive point. The physical scales are

$$\psi_1(k) = a^2/k^2, \quad \psi_2(k) = \max\{\max(b, c)^2/k^2, bc/k\} \quad (bc > 0),$$

with zero widths removed at the boundary. There is no single smooth acquired manifold at the crossing or a single ambient-dimension exponent. Theorem 5.2 applies with constants uniform through all these changes. Smooth nonalgebraic bi-Lipschitz distortions of the two charts yield the same conclusion, with their uniform metric constants. Thus the extension is not a statement about a preassigned Hilbert-space rank cut.

6. CONSEQUENCES, DISTINCTIONS, AND PROOF BOUNDARIES

The general conclusion has three operationally different ingredients: the full-history acquisition risk B_N , the acquired geometric allocation Ψ_N , and the forced transport operator in (8). They originate in the same raw experiment but are not interchangeable. A formal parameter dimension controls none of them by itself. In the singular sensor, for example, the raw target may carry coordinates killed by the report map, the absorbed law may mix dimensions, and its total mass is only $1 - (1 - s)^N$.

The two realizations verify different causal mechanisms. The filter has positive continuous likelihoods, repeated Bayesian normalization, and contraction in log-odds; its acquired density is established by an actual report Jacobian. The singular sensor has an absorbing reveal, exact holds of arbitrarily long duration, and an acquired measure with intersections and atoms; its transport is a disjoint indicator identity. Neither calculation is a premise of Theorem 2.1. The main proof consists of the predictive realization, actual-mass lower estimate, global valid cover, and acquisition-weighted transport argument, followed by common-task morphism composition.

6.1. Relation to existing theory. The checkpoint identity is the conditional-mean projection underlying quantization, not a new alternative to it. The anisotropic box proof is included to expose its constants through vanishing physical widths and to accommodate actual component weights and discarded components. The general study of quantization of singular probabilities is much broader than this box lemma [4]; local error estimates for Ahlfors–David measures provide another developed direction [8]. We do not claim that the present paper supersedes those results.

The difference from numerical filter quantization [6, 7, 3] is the simultaneous lower bound for the same acquired task and the explicit separation of retained labels, workspace, program data, clock,

and raw acquisition. The innovation argument does not charge another rounding on an exactly represented hold. The difference from unforced random-iteration convergence [2] is the weighted second-moment estimate for the forcing introduced by a finite machine. The experiment-comparison argument [1] remains a common-loss argument; subtracting two different oracles would destroy it.

6.2. Necessary distinctions illustrated by exact examples. First, the support $[0, 1]$ equipped with a point mass and the same envelope equipped with a uniform law have respectively zero one-point distortion and strictly positive finite-label distortion. Hence a reachability envelope is not an acquired law. More generally a component of probability w contributes with weight w , even if its geometric diameter is large.

Second, an identity update has no need for repeated quantization. If the initial state has already been replaced by a representative, the same representative can be retained indefinitely. The bound Nr obtained by blindly injecting a covering radius at every hold is not an inherent resource requirement. The acquisition-weighted certificate corrects precisely this failure of accounting.

Third, a statistical garbling can increase memory needed to match its *own* oracle. Observing a fair hidden bit exactly produces only posterior probabilities zero and one. Adding independent Gaussian noise to this observation produces a continuously varying posterior probability. Two labels reproduce the first oracle exactly, but no finite number reproduces the latter continuously distributed oracle exactly under square score. This is compatible with data processing because the baselines differ.

Fourth, let $P_N = \delta_0$ and $Q_N = (1 - e^{-Nc})\delta_0 + e^{-Nc}\delta_1$, $c > 0$. Their total variation distance tends to zero, but the exponential rates at the point one are infinite and c , respectively. The common-risk defect of this paper is therefore not a certificate of large-deviation equivalence.

Finally, a hidden mode or an internal simulator register is not free because its name is omitted from a state vector. Independent pairs of retained states have the product cardinality in Theorem 4.1. Conversely that product is a constructive upper bound, not a theorem that every simulator requires it. An exact label machine and a finite-bit numerical implementation are also different objects: the latter retains the nonzero calibration, workspace, and readout terms of (20). Exact-arithmetic zero distortion is not a promise of a finite-precision physical output with zero error.

6.3. Scope of the mathematical conclusion. The main theorem is a conditional but quantitative class theorem: its certificates do not contain optimal risks, and they are completely checked for the two raw-kernel classes above. It establishes finite-horizon Bayesian prediction curves under fixed exploration, and common-task causal risk transfer over whatever strategy class has a uniform simulator certificate. It does not prove an unrestricted optimal exploration theorem, a parameter-uniform predictive quotient for every prior, or the learning of an unknown kernel from an unpaid pilot. The intersecting-strata result is uniform in distances and widths for finitely many uniformly bi-Lipschitz acquired charts, at the stated budget threshold. General singular laws without this geometry require new actual-mass estimates, not a formal change of notation.

The remaining extensions are specific mathematical questions. One is to derive acquisition-weighted moment bounds for recurrent nonuniform filters that may expand during active updates, beyond the exact-hold and contractive-active mechanisms proved here. Another is to derive uniform charts and optimal calibration acquisition costs from unknown kernel classes. A third is to match the complete resource vector, including growing controller and simulator states, without assuming constant overhead. Infinite-horizon path-law equivalence, large deviations, and unbounded-generator closure require their own stronger certificates. None is inferred from the finite-time theorem or delegated to a realization as an unproved foundation.

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